Practical aspects of training / validation

Predict Who Will Repay Their Loans

- A lender hires you to investigate who will repay their loans
  - You are given complete files on 100 applicants
  - 5 defaulted within 3 years

Image credit debt.org

A Surprising Finding

- All 5 people who defaulted wore blue shirts during interviews
- Your model may find this strong signal as well

Training Error and Generalization Error

- Training error (bias): model error on the training data
- Generalization error (variance): model error on new data
- Example: practice a future exam with past exams
  - Doing well on past exams (training error) doesn't guarantee a good score on the future exam (generalization error)
  - Student A gets 0 error on past exams by rote learning
  - Student B understands the reasons for given answers